

Improving Driver Supply in Uber

Solving the supply-side marketplace problem to reduce wait times and improve rider experience



Problem Statement: Improving Driver Supply

The Problem

Insufficient driver supply during peak hours leads to surge pricing and long wait times	High driver churn rate (30-40% annually) creates persistent supply gaps	Uneven geographic distribution leaves certain zones underserved
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Why It Matters

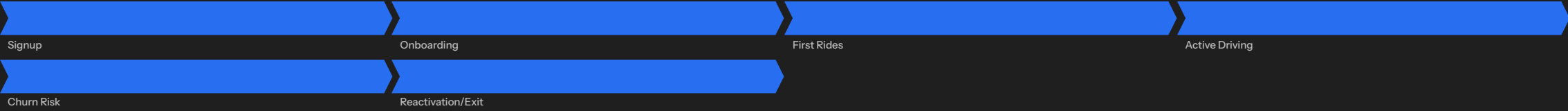
For Riders	For Drivers	For Business
8-12 min avg wait times - Surge pricing frustration - App abandonment	- Inconsistent earnings - Low platform loyalty - Burnout from long hours	- Lost ride requests - Higher CAC for new drivers - Market share loss to competitors

Key Metrics Impacted

Driver Online Hours per Week — currently 18 hrs avg , target: 25+ hrs	Supply-Demand Ratio — currently 0.7 during peak, target: 0.9+
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Driver Lifecycle Analysis

Driver Lifecycle Flow



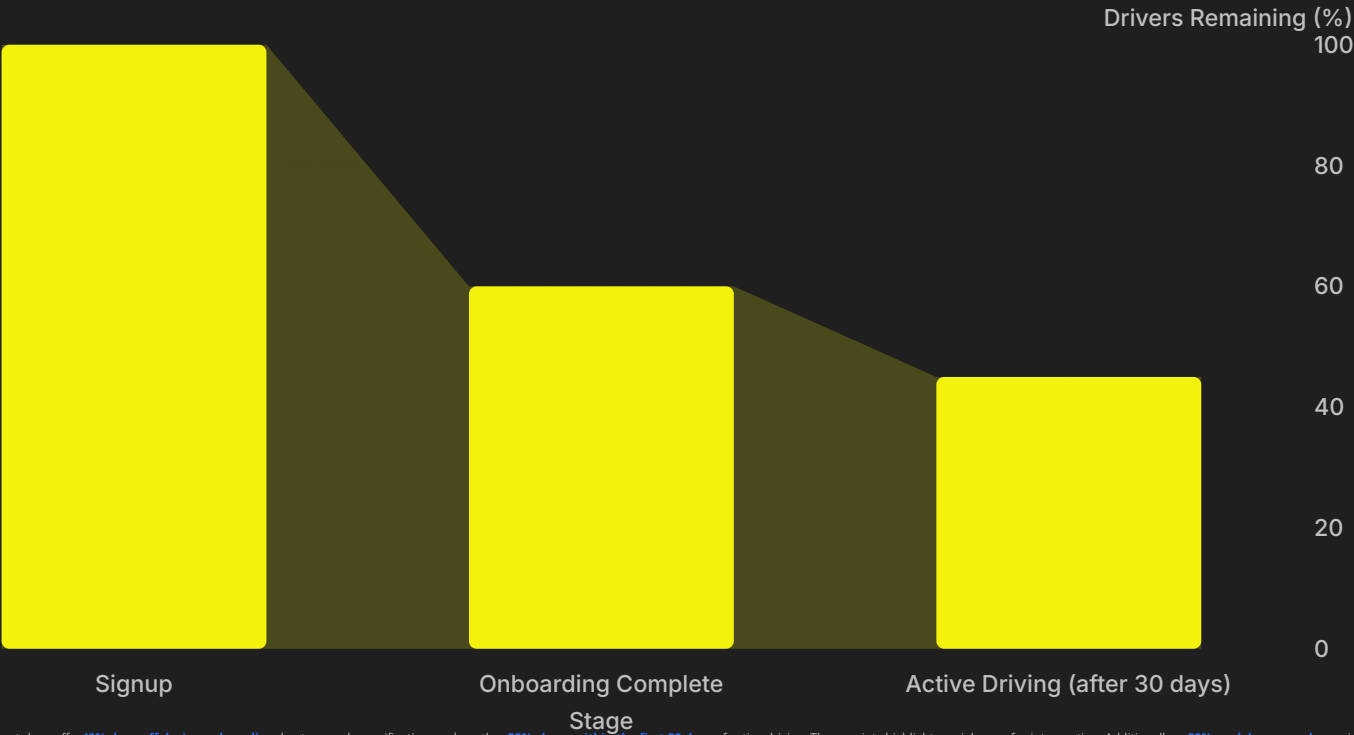
Where Problems Occur

Stage	Pain Point
Onboarding	Complex document verification (3-5 days)
First Rides	No guidance on high-demand zones/times
Active Driving	Unpredictable earnings, dead miles
Churn Risk	No early warning or retention interventions

Critical Drop-off Points

45%

Total conversion rate



Analysis shows significant drop-offs: 40% drop-off during onboarding due to complex verification, and another 25% churn within the first 30 days of active driving. These points highlight crucial areas for intervention. Additionally, a 30% peak hour supply gap indicates ongoing challenges in driver availability.

Key Pain Points for Drivers



Earnings Unpredictability

- Drivers can't forecast daily/weekly income
- No visibility into demand patterns
- Dead miles between rides eat into profits



Slow Onboarding

- Document verification takes 3-5 days
- Multiple app visits required to complete signup
- No progress tracking or proactive communication



Lack of Demand Visibility

- Drivers rely on intuition for positioning
- Surge zones appear too late to act on
- No personalized recommendations for optimal hours



Weak Driver Engagement

- No gamification or progression system
- Limited recognition for high performers
- Support issues take 24-48 hrs to resolve

Proposed Solutions



Earnings Estimator & Planner

- **Weekly earnings forecast** based on historical patterns
- "Best hours to drive" **personalized recommendations**
- **Real-time earnings tracker** with goal progress



Express Onboarding

- **AI-powered instant document verification**
- **Single-session signup flow** (under 15 mins)
- **First-ride guarantee** within 24 hrs of approval

Smart Demand Heatmaps

- **Predictive surge alerts** 30 mins in advance
- Personalized zone recommendations based on **driver location**
- **"Navigate to demand"** one-tap routing



Driver Rewards Program

- **Tiered status system** (Bronze → Platinum)
- **Weekly challenges** with bonus incentives
- **Priority support** for top-tier drivers

Experiments & Validation

A/B Testing Plan

Experiment	Control	Variant	Duration
Earnings Estimator	No forecast	Weekly planner	4 weeks
Express Onboarding	Current flow	AI verification	3 weeks
Predictive Heatmaps	Current maps	30-min predictions	4 weeks
Rewards Program	No program	Tiered rewards	6 weeks

Validation Approach

Qualitative

Driver interviews (n=30) across experience levels

Quantitative

Cohort analysis on driver retention curves

Operational

Supply-demand ratio monitoring by zone

Success Criteria

15% increase in weekly online hours

20% reduction in onboarding drop-off

No increase in driver support tickets

Success Metrics

Primary Metrics

Metric	Current	Target	Rationale
Avg Online Hours/Week	18 hrs	25 hrs	Core supply metric
Driver Retention (90-day)	60%	75%	Reduces acquisition cost
Peak Hour Supply Ratio	0.7	0.9	Directly impacts rider ETA

Secondary Metrics

Metric	Current	Target
Onboarding Completion Rate	60%	80%
Avg Rider Wait Time	8 min	5 min
Driver NPS	32	50
Surge Frequency	35% of rides	20% of rides

Guardrail Metrics

- Rider cancellation rate (should not increase)
- Driver earnings per hour (should not decrease)
- Safety incident rate (should not increase)

Expected Impact & Key Takeaways

Expected Business Impact

Metric	Improvement	Business Value
Supply Increase	+40% online hours	Reduced surge, better rider ETA
Retention	+15% at 90 days	\$50M+ savings in driver acquisition
Wait Time	-3 minutes avg	Higher rider conversion & retention

Key Product Insights

01

Supply drives demand

Solving driver problems directly improves rider experience

03

Onboarding is make-or-break

First 7 days determine long-term retention

02

Predictability > raw earnings

Drivers value income stability over occasional bonuses

04

Recognition matters

Status and progression keep drivers engaged beyond financial incentives

Next Steps

- Prioritize Express Onboarding (fastest time-to-impact)
- Build earnings prediction ML model
- Design driver segmentation framework for personalization